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SURVEY PAPER

Unmanned aerial vehicles using machine learning for autonomous flight; state-of-the-art

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ABSTRACT
In recent years, since researchers began to study on Unmanned Aerial Vehicles (UAVs), UAVs have been integrated into today's everyday life, including civilian area and military area. Many researchers have tried to make use of UAVs as an ideal platform for inspection, delivery, surveillance, and so on. In particular, machine learning has been applied to UAVs for autonomous flight that enables UAVs do designated task more efficiently. In this paper, we review the history and the classification of machine learning, and discuss the state-of-the-art machine learning that has been applied to UAVs for autonomous flight. We provide control strategies including parameter tuning, adaptive control for uncertain environment, and real-time path planning, and object recognition that have been described in the literature.

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1. Introduction

Unmanned aerial vehicles (UAVs) have become an attractive area of research increasingly in recent years [1–7]. Many researchers have been studying on UAVs to utilize as an ideal platform for civilian tasks or military tasks, such as inspection, delivery, reconnaissance, or surveillance like the mobile robots based on autonomous navigation, real-time path planning, and object recognition [8–23]. However, UAVs have some challenges for autonomous flight, such as control strategy including parameter tuning, adaptive control, real-time path planning, and object recognition under uncertain environments. In previous studies, some approaches such as a negotiation approach, a heuristic approach, and a graph theory approach were proposed to solve it [24–26]. However, these approaches still had difficulties with sensors, system dynamics, qualities, and so on. In recent years, machine learning has become an attractive approach to overcome these challenges in UAVs for autonomous flight. Machine learning enables UAVs to recognize patterns or predict from data without designed programming for autonomous flight.

Efe [50], Wang et al. [51], Gao et al. [52], Santos et al. [53,54], and Jardine et al. [55] tried to tune a parameter using machine learning for the controller in UAVs to adapt a new environment quickly. Kurnaz et al. [60], Chowdhary et al. [61], Lin et al. [65], Punjani et al. [66], Bansal et al. [68], Shin et al. [69], Bou-Ammar et al. [71], and Lee et al. [72] tried to apply adaptive control algorithms using machine learning for autonomous flight in UAVs. Geramifard et al. [78], Junell et al. [81], Zhang et al. [79], Xu et al. [21], Kan et al. [82], and Loquercio et al. [83] proposed machine learning algorithms for real-time path planning in UAVs to flight autonomously. Lei et al. [90], Ross et al. [91], Junell et al. [81], Yang et al. [92], Kahn et al. [93], and Zhang et al. [94] tried to avoid collision in real time against obstacles or other aerial vehicles using machine learning algorithms, and Cherian et al. [95], Guo et al. [97], Perez-Ortiz et al. [99], Rebetez et al. [100], Giusti et al. [101], Smolyanskiy et al. [102], and Bartak et al. [104] proposed machine learning algorithms to recognize object for the task of UAVs such as altitude of UAVs, landing site, and so on.

Although there have been remarkable research accomplishments of UAV flight [27–30] and indoor navigation [31–36], we will focus our review on machine learning combined autonomous flight, including control strategies and object recognition. We attempt to cover the research areas described above, particularly featuring control strategies including parameter tuning, adaptive control under uncertain environments, and real-time path planning, and object recognition including collision avoidance and object recognition on task. When it

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is available, we describe the results of any evaluation of their proposed machine learning algorithms.

In this paper, we begin with a brief background on machine learning in order to describe key concept of machine learning applied in UAVs for autonomous flight. Then, we move on to reviewing the previous studies. Lastly, we present a discussion of this reviewing and future work for autonomous flight in UAVs based on machine learning.

2. Machine learning

In 1950, Alan Turing introduced the concept of Artificial Intelligence (AI) after the first computers were developed [37,38]. He proposed the Turing Test, which attempted to distinguish machine from people through several questions and answers. If the computer passed the test, it was considered as 'intelligence' like human. Based on the Turing Test, some researchers studied on AI, then AI was established as an academic discipline in 1956 [39]. AI was defined as intelligent agent to perceive variable environment and decide what to do with learning and problem solving. However, even though many researchers tried to apply AI to different research areas over 60 years, it was not remarkably developed until 21st century, because there were some challenges to perceive variable environment as a limitation of computational capability. In 2000s, AI has been noticeably developed with the popularization of the Internet, sensors, big data, interconnection and fusion of data and knowledge and so on [40]. Nowadays, AI has been studied in many research areas, such as natural language processing, knowledge representation, automated reasoning, and machine learning [41,42]. Among them, machine learning is one of key research areas for AI. Many researchers have been trying to recognize patterns or make predictions from data using machine learning without programming the computer. In particular, machine learning is being studied in unmanned aerial vehicles (UAVs) for autonomous flight or intelligent behavior. Machine learning enables UAVs to solve various challenges such as intelligent control strategy and object recognition.

Machine learning is classified into three categories: supervised learning, unsupervised learning, and reinforcement learning [41,43].

Supervised Learning (SL) is the most common form of machine learning. In supervised learning, the agent observes some input-output pairs, and learns a function that maps output from input [41]. The agent is interested in labeling the output with the condition of knowing the input. For an input x, if the output is one of a discrete number of possible 'classes', this is called a classification problem. If the output is continuous, it is called a regression problem [44].

Unsupervised learning (UL) has no labeled output. Instead, the agent focuses on observing the world and learning patterns without being given labeled data. It aims to find the hidden pattern, structure, or features embedded with the data [43]. A representative example of unsupervised learning is clustering. By simply eyeballing the data, the agent is able to see that there are several apparent clusters, then classify each data into its corresponding cluster [44].

Reinforcement learning (RL) is interested in finding the optimal policy which is learned from feedback. This kind of feedback is called reward or reinforcement [41]. It repeats trial and error until the agent chooses which of the actions is appropriate to maximize the expected total reward. Therefore, the task of reinforcement learning is to learn an optimal policy for the environment by using observed rewards [41]. Classical reinforcement learning approaches are based on the assumption that it has Markov Decision Process (MDP) consisting of the set of states S, set of actions A, the rewards R, and transition probabilities T that capture the dynamics of a system. $T(s', a, s) = P(s'|s, a)$ describes the effects of actions on the state. The Markov property requires that the next state s' and the reward only depend on the previous state s and action a, and not on additional information about the past states or actions. Therefore, a state is considered as a sufficient statistic for predicting the future, rendering previous observations irrelevant [45–47].

Deep learning (DL) is also a typical machine learning method based on the intelligent behavior of the human brain. To explain complex systems, DL algorithms use neural networks (NN) that consist of hierarchy of simple neurons like the human brain. DL algorithms can be classified into supervised, unsupervised, and reinforcement learning according to their characteristics [48,49].

3. Machine learning applied to UAVs for autonomous flight

Machine learning has contributed to broad fields for autonomous flight of UAVs. We particularly discuss the state-of-the-art control strategies and object recognition for autonomous flight of UAVs. Figure 1 shows the timeline of previous studies that are covered below.

3.1. Control strategies

In UAVs, control strategy is a key to autonomous flight despite highly nonlinear dynamics and uncertainty in real world. As a result, many researchers have been studying on control strategies enhanced by machine learning.

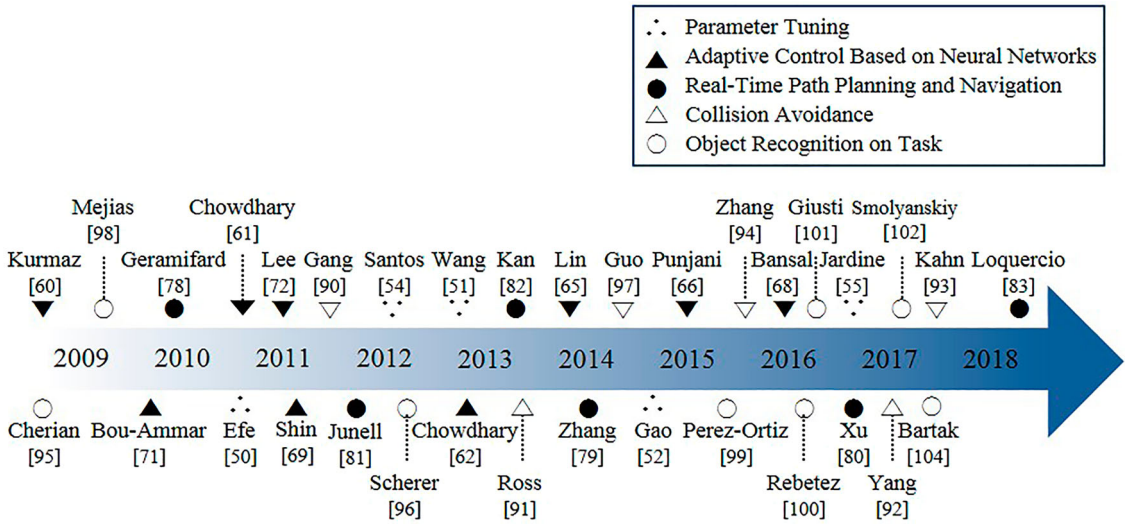

Figure 1. Timeline of previous machine learning related studies for UAVs.

We divide research areas into three categories: parameter tuning, adaptive control, and real-time path planning and navigation.

3.1.1. Parameter tuning

Parameter tuning for the objective function in the control system is the simplest application of machine learning. In real time, parameter tuning is quite important for the controller in UAVs to adapt a new environment quickly.

There were several studies on parameter tuning using machine learning algorithms to find optimal control gains of PID controller in UAVs. Efe [50] proposed that coefficient of an approximator which approximates the response of the $PI^{\lambda}D^{\mu}$ controller can be obtained by training the feedforward neural networks (FNNs) [50]. Fractional order $PI^{\lambda}D^{\mu}$ controller is a generalization of the integer order PID controller exploiting the richness offered by the non-integer orders of the Laplace variable s. The simulation results demonstrated similar response of the $PI^{\lambda}D^{\mu}$ controller and much less computation. However, it is required to test more complex models because the demonstration is restricted to simple linear models. Wang et al. [51] introduced an intelligent PID control based on Radial Basis Function (RBF) NN that enables the PID controller adapt different environment in UAVs for autonomous flight by optimizing control parameters [51]. RBF NN is a three-layer feed forward network, including an input layer, a single hidden layer and an output layer. In the RBF NN, they utilized radial basis Gaussian function as the network activation function of the hidden layer. Through the simulation, RBF NN control algorithm achieved optimization of the PID parameters for pitch angle control more efficiently than traditional PID control. For future progress, the algorithm needs to be applied to roll and yaw angles, and control the angles together at the same time. Gao et al. [52] also demonstrated RBF NN-based PID control algorithm that overcomes nonlinearity and parameter uncertainty of the PID method [52]. This algorithm is capable of self-adaption and online learning. Unlike the controller of [51], they designed both attitude and position controller of quadrotor to track the predicted trajectory. They demonstrated RBF NN-based PID algorithm recovered faster than the conventional PID algorithm from a wind disturbance, and reduced 17% of overshoot from the PID. However, it did not show better performance in tracking and anti-disturbance than PID algorithm. It will be interesting to overcome inferiority of proposed method.

Santos et al. [53,54] presented optimization of quadcopter attitude and path tracking controller using learning automata (LA) algorithm for adjusting the parameters of the PID controller [53,54]. Learning automata, one of reinforcement learning method, is an adaptive decision-making method that learns the optimal action out of a set of actions through repeated interactions with a random environment. Using LA, the parameters of controllers were adjusted for a nonlinear UAV, taking into account the nonlinear disturbances. The simulations executed nonlinear tracking and flip maneuver, and the experiments conducted position tracking performance. The simulation results showed better performance compared with the same controller adjusted using the conventional method, particularly in presence of external disturbance like wind and ground effect. The experiments presented similar performance with simulated tests in tracking performance. As simulation results, it is necessary to test aggressive maneuvers in proposed experimental framework. Jardine et al. [55] also utilized

LA to choose the optimal weighting parameters of objective function when planning quadcopter trajectory based on model predictive control (MPC) [55]. MPC-based techniques provide efficient energy consumption of the autonomous vehicles, however, it requires high computational demands of the optimization step. They demonstrated that LA improved tracking error performance by selecting weighted cost function over time. For future work, the tuning techniques will be investigated in actual physical system of a quadcopter.

3.1.2. Adaptive control based on neural networks

There have been various applications of adaptive control algorithms based on neural networks (NNs) for autonomous flight in UAVs. The increasing interest in adaptive control stems from its ability to handle changes in system dynamics and to mitigate uncertainty. Based on the intelligent behavior of the human brain, NN consists of many simple, connected processors called neurons, each producing a sequence of real-valued activations [56–59]. This yields the great advantages in high-dimensional data representation and processing, therefore, NN has been widely used to solve complicated adaptive control problems of UAVs for autonomous flight [43].

Kurnaz et al. [60] developed an adaptive neuro-fuzzy inference system (ANFIS) based controller, that is a five layered feedforward NN structure for the autonomous operation of UAVs [60]. In this work, a hybrid learning algorithm was utilized for updating ANFIS network parameters. During the forward pass, least square error-based training was utilized for the output parameters of the network while the input parameters were kept constant. During the backward pass, the output parameters were kept constant and the parameter set of input was updated using gradient descent method. This system was applied to steep turn maneuver for UAVs as a reference trajectory through simulation. However, the ANFIS type controller demonstrated inferior performance compared with the well tuned PI and fuzzy type of controller, and resulted in unstable performance for some flight conditions.

Chowdhary and Johnson [61] introduced concurrent-learning (CL) adaptive controller that was aimed at improving the convergence of adaptive weights in model reference adaptive control (MRAC). They utilized a single hidden-layer NN as the adaptive element, and selected and stored data from online was utilized concurrently for adaptation to overcome the rank 1 limitation of the NN weight adaptation law. They demonstrated the improved tracking performance of the adaptive controller with concurrent-learning compared with the controller without concurrent-learning experimentally using a rotorcraft UAV. Chowdhary et al. [62] applied the concurrent learning to the problem of the control transfer between systems that have similar control structure. The problem of control-transfer was defined as the task of transferring controllers from one UAV to another without sacrificing tracking performance. They utilized two learning-based MRAC laws: concurrent learning MRAC, and budgeted kernel restructuring concurrent learning (BKR-CL). In particular, BKR-CL method was firstly presented as a budged nonparametric adaptive control approach in this paper.

Lin et al. [65] proposed an intelligent control system using a recurrent wavelet neural network (RWNN). The recurrent NNs (RNNs) have the capabilities superior to the feedforward NN, such as the dynamic response and information storing ability. Recently, wavelets have been combined with the NN to create wavelet neural networks (WNNs) for achieving better approximation performance than NNs. RWNNs combine the properties of dynamics response of RNN and the fast convergence of WNN [63,64]. This study demonstrated the RWWN-based control system achieves favorable tracking performance for the motion control of UAV even for effort deterioration and crosswind disturbance [65]. However, the experimental tests for actual fixed-wing UAVs are needed to verify tracking performance of this work.

Punjani and Abbeel [66] introduced Rectified Linear Unit (ReLU) Network Model for aerobatic maneuver of the helicopter, providing details about parameter initialization and optimization methods. The aerobatic helicopter is known as difficult to both model and control due to its complex maneuvers, and it has only been successfully flown under autonomous control using apprenticeship learning, which leverages expert demonstrations to efficiently learn good controllers for tasks being demonstrated by an expert [67]. They utilized real-world data collected by an expert pilot from the Stanford Autonomous Helicopter Project. They demonstrated that the ReLU Network Model improved RMS acceleration prediction by 58% over prior state-of-the-art for dynamics modeling of the aerobatic helicopter even though this model didn't enable autonomous control of the helicopter through aerobatic maneuvers.

Bansal et al. [68] focused on a NN based dynamic model which can be employed to synthesize control rather than improving the control performance for specific trajectories [68]. Learning the dynamics of UAVs using a NN can be utilized to track the trajectories the NN model was not trained on. Their experiments indicated that even simple NNs such as feedforward networks were also able to not only have a good capabilities for generalizations but also learn the dynamics of a quadrotor for good accuracy. As a result, the learned dynamics

could be utilized effectively to control the system. For future improvements, it will be interesting to utilize both the known information about the system as well as the generalization capabilities of NNs.

On the other hand, adaptive control algorithms have been reinforced using different machine learning approaches rather than NNs, such as support vector machine (SVM) and support vector regression (SVR). Support vector machine and SVR have been also studied for adaptive control of UAVs. SVR generates global solutions because SVR basically solves quadratic programming problems. Shin et al. [69] applied SVR-based control to input-output feedback linearization technique [69]. In particular, two SVR machines are trained offline. The first is called the inversion SVR that is designed for training the feedback-linearized inverse dynamic model. The second is compensation SVR that estimates the output derivative. An online adaptation rule is applied for compensation SVR to handle the unexpected nonlinearities and uncertainties. The simulation results of tracking performance showed the superiority of the SVR-based algorithm compared with the NN-based adaptive control. SVR-based adaptive control can be a good alternative to the NN-based adaptive control when unexpected inputs are involved. In addition, reinforcement learning has been applied to solve control challenges [45,70]. Bou-Ammar et al. [71] presented a comparison between two control approaches, the nonlinear feedback controller and the controller learned using MDP and fitted value iteration (FVI) method to approximate the value function for a quadrotor UAV [71]. The control algorithm without RL showed better performance but it needed deeper knowledge of dynamic behavior. This is known as one of the key advantages of reinforcement learning when applying the control strategies. For future improvements, non-parametric approximation technique like a wavelet network can be further studied and tested on the FVI algorithm. Lee et al. [72] adopted a model-free discrete linear quadratic control instead of PID control for unmanned helicopters. It was tied with Q-learning algorithm, which trained the feedback gain with small errors [72]. For the simulation, the Cart-Pole simulation was carried out, which is well known simple dynamic system. It will be necessary to apply this approach to more complicated maneuver.

3.1.3. Real-time path planning and navigation

Real-time path planning and navigation have been considered as a fundamental and necessary task in UAVs for autonomous flight under diverse civil applications, such as surveillance, rescue, guidance of air and so on. These technologies are particularly important for multiple UAVs to provide safe and optimal paths. Recently, many researchers have studied path planning and formation flight techniques [11,73,74]. Below, we present path planning and navigation methods enhanced by machine learning for both single and multiple UAVs.

Cooperative planners have been developed to compute non-colliding paths for multi-agent UAVs, yet it is based on simple models with inaccuracies. Geramifard et al. studied on the intelligent Cooperative Control Architecture (iCCA) as a framework for combining cooperative planners and reinforcement learning techniques [75–78]. iCCA improved the policy of cooperative planner, while reduced the risk and sample complexity of the learner. Empirical results of mission planning scenario showed iCCA boosted the performance of fixed planners and reduced the risk and sample complexity of the learner.

Zhang et al. [79] presented cooperative and geometric learning algorithm (CGLA) that is designed for path planning based on cooperation of multiple UAVs [79]. In CGLA, the parameters were trained to balance between the collision avoidance and economy of the paths. The individual weight matrix and cost matrix were proposed for an efficient path planning of both single and multiple UAVs. Specifically, the individual weight matrix was calculated and adaptively updated based on the geometric distance and risk information shared among UAVs, and the cost matrix was calculated to prove the optimal path from a starting point to a target point. The simulation results validated the effectiveness and feasibility of CGLA for safe navigation of multiple UAVs. However, it is required to utilize coarse and fine grid to design fast path planning algorithm in the proposed framework, and apply the algorithm to an actual UAV.

Xu et al. [21] proposed reinforcement learning-based optimal flocking control and network co-design for the networked multi-UAV system [80]. Networked multi-UAV is able to share the local information between UAVs to avoid collisions and construct an effective path control in real-time, whereas the network uncertainties is also able to bring serious challenges. To overcome network imperfections and inaccurate UAV dynamics, they studied on a novel online finite horizon optimal flocking control and network co-design for networked multi-UAV with network constraint to combine the control and network aspects effectively. To obtain the ideal optimal co-design, two actor NNs were developed to estimate the optimal system control signals and wireless communication power control inputs efficiently for distributed multi-UAV. However, there is a slight overshoot at the beginning because the proposed design needs some time to tune the NNs.

Junell et al. [81] focused on path planning for a single UAV [81]. Junell et al. [81] studied reinforcement

learning to find an optimal mission path of an UAV [81]. RL can be an appropriate approach for the high level control decision like optimization of a path when given an unknown environment. The test was set up to demonstrate that a quadrotor can autonomously visit disaster sites and take pictures as many as possible (reward states) within a 6×6 grid world. Once the quadrotor learned a value function in simulation, flight tests were implemented using the simulation results. This approach can reduce huge iterations on the flight tests. However, this work is limited to the discrete and unchangeable domain.

Kan et al. [82] proposed a terrain-based navigation for UAVs using extreme learning machine (ELM) algorithm [82]. The terrain-based navigation enables UAVs to ascertain its positional information by capturing terrain data from the onboard sensor when global positioning system shuts down or operates abnormally. ELM algorithm is similar to a single layer feedforward network, which is used to reduce training time of NNs. Compared with other algorithms such as Delaunay triangulation interpolation and back propagation, test results verified ELM algorithm required much shorter time when computing the multi-resolution terrain and the terrain with a large size. However, reconfigurable and context independent hardware will be required for the test improvements. Loquercio et al. [83] focused on civilian drones, which are expected to be used monitoring, delivery, or surveillance in urban environments [83]. They proposed a Convolutional Neural Networks (CNNs) called DroNet to navigate autonomously through the streets of a city. In contrast to 'map-localize-plan' methods, DroNet is a data-driven approach, which requires a huge amount of data. They collected data from cars and bicycles, minimizing the risk that UAVs, involve for other vehicles or pedestrians in the streets. The main advantage of this approach is that it does not need a map of the environment or build it online. The limitations are that this approach is constrained with resources in the streets of a city, and that the agile dynamic of a drone is not fully exploited, as it is common in other CNN-based controllers.

3.2. Object recognition

Object recognition is also one of key challenges for UAVs. It is required for UAVs to recognize object to avoid collision in real time against obstacles and other aerial vehicles. UAVs try to recognize many kinds of objects according to the mission of the UAV, such as agriculture, rescue, surveillance, and so on. In addition, UAVs try to recognize object for checking altitude and landing area. There have been mainly two kinds of approaches to recognize objects; cooperative approach and non-cooperated approach. Cooperative approach is to communicate each other using a transponder. Non-cooperated approach is to detect object and decide what to do using acquired or trained dataset. Recently, many researchers have been trying to recognize objects using machine learning.

3.2.1. Collision avoidance

There were many impressive results for UAVs to avoid collision using laser range finders or Microsoft Kinect cameras [84–87]. Vandapel et al. [84] tried to avoid collision in 3D space using lidars, and Bachrach et al. [85] and Bry et al. [86] showed obstacle avoidance in unknown room using scanning lidars for Simultaneous Localization and Mapping (SLAM) [84–86]. Bachrach et al. [87] used a Microsoft Kinetic cameras in unknown room [87]. However, these research were required to improve problems of power supply, payload, UAVs cost and the reliability of localization. As a result, Achtelik et al. [88], Wendel et al. [89], Fraundorfer et al. [28] created maps by a single camera, or stereo cameras [28,88,89]. Achtelik et al. [88] created sparse maps of the environment using a single, cheap camera [88]. Wendel et al. [89] created dense maps of the environment using a camera [89]. Fraundorfer et al. [28] demonstrated accurate depth estimation and localization using stereo cameras [28]. However, even though the algorithms were reasonably fast to avoid collision, they were still too computationally expensive for UAVs flight. Various machine learning techniques for collision avoidance come into the picture. Lei et al. [90] proposed a restructured Q-value learning algorithm to avoid collision for UAVs [90]. They restructured Q-value learning algorithm into a dynamic reinforcement learning algorithm. They proposed scoring function to analyze the threat of collision. However, this approach has limitations with undefined conditions. It is required to design an algorithm for the calculation of reward value in order to more complex and powerful in undefined conditions. Ross et al. [91] adapted a imitation learning strategy to train reactive heading policies acquired from a human pilot [91]. They succeeded in flying MAVs at low altitude using a cheap camera through natural forest environments. They avoided more than 680 trees over 3.4 km during flights. However, narrow field of view of the camera often caused failures of the reactive approach. Junell et al. [81] proposed the setup and a high level reinforcement learning algorithm for UAVs to avoid collision [81]. They demonstrated real flight test with a quadrotor in unknown environment to take a picture about a disaster site. The quadrotor could learn a converged, near optimal value function with many iteration. However, this approach is required not only to demonstrate under unknown environments, but also to design policy selection in Q-learning algorithm

for selecting policy automatically and reducing iteration. Yang et al. [92] tried to avoid collision using deep networks based intermediate perception [92]. They proposed a new approach to predict a trajectory for the quadrotor from images. They labelled trajectory from image automatically and create a large dataset. Then, they predicted the trajectory using a 3D model based on CNNs. Simulation results of obstacle avoidance demonstrated 20% more accuracy than the quadrotor controlled by the operator. This approach is required to show the reliability of dataset and to reduce training from it. Moreover, it is required to estimate accurate states and to predict more variable trajectories. Kahn et al. [93] proposed an uncertainty aware reinforcement learning algorithm that estimates the probability of the collision with a statistical estimate of uncertainty for high-speed UAVs [93]. They formulated an uncertainty dependent cost function, and utilized bootstrapped NNs using dropout with high-bandwidth cameras. They demonstrated their proposed reinforcement learning algorithm with the quadrotors at high-speed. The result showed that their proposed reinforcement learning algorithm with uncertainty had fewer collision than without uncertainty. However, this approach is required to estimate uncertainty more accurately and to demonstrate it in undefined environment for robustness.

Zhang et al. [94] combined MPC with reinforcement learning [94]. They proposed MPC to generate data at training time, which is used to train a deep neural network policy. They evaluated their method by showing two scenarios of obstacles avoidance. At test phase, the trained DNN policy successfully controlled UAVs without knowing full states. However, they had some limitations that full state observations were required at training phase, and not all aerial maneuvers can be learned in this instrumented training setup.

3.2.2. Object recognition on task

In this section, we explore object recognition according to a variety of missions of UAVs. Cherian et al. [95] proposed a semi-supervised machine learning (SSL) algorithm to estimate of the altitude of UAVs using top-down aerial images with a single camera [95]. They tried to learn the mapping between the information in top-down aerial images to the altitude through regression of sparse basis set against a training set of altitudes. A spatio-temporal Markov Random Field was modeled to demonstrate their algorithm including a real helicopter. However, their demonstration was only done at low altitudes and low speeds and had limitation in the textureless ground surface. Scherer et al. [96] developed a SVM for navigating and mapping obstacles along a river by a low-flying MAV [96]. They succeeded in traveling 2 km along a river, and showing that mapping the river course, width, and canopy height using lightweight sensors equipped on a MAV.

Guo et al. [97] proposed the machine learning algorithm for UAVs to detect landing site using the Gaussian Mixture Model and the Support Vector Machine [97]. Support vector kernels including RBF Kernel, polynomial kernel, the order of RBF and polynomial kernel, and linear, and the number of Gaussian mixtures were analyzed as a range of learning parameters. They introduced a modified footprint operator for the extraction of feature. They demonstrated an automatic visual image-based aerial image classification based on machine learning algorithm and compared against the baseline Artificial Neural Network system was proposed by Mejias et al. [98]. A 35.93% improvement and more stable performance were presented. However, it was required more sampling window sizes and more advanced features. Perez-Ortiz et al. [99] proposed the machine learning algorithm for precision agriculture by UAVs from sunflower and maize crops [99]. They tried to specify the spectral similarity of crop and weed pixels in early growth steps using object-based image analysis (OBIA) based on machine learning algorithm. They combined with the technology of pattern and feature selection. They chose a set of training patterns through clustering approaches for the classification method and obtained the best discriminating features from a statistical set and variable environment. They presented that the expert needs only the data acquisition process using UAVs and the data labeling when it was provided by their proposed machine learning algorithm. However, this approach is required to design model to reduce computation and to demonstrate variable environments. Rebetez et al. [100] proposed the machine learning algorithm using a hybrid deep NN for crop classification from high-resolution UAV imagery [100]. They tried to recognize the types of crop from aerial image of UAVs. They presented a new hybrid NN architecture using histograms and convolutional units (CNN). They utilized the raw pixel value for CNN and RGB histograms for a dense side. The outputs of the CNN and RGB histograms were merged into a dense layer for the classification. The result of the classification was improved better than either model alone. However, this approach has limitation with predefined scenario. This approach cannot apply into different environments and dataset. As a result, this approach is required to design an algorithm to learn under different environments with variable dataset.

Giusti et al. [101] and Smolyanskiy et al. [102] studied the problem of perceiving forest and mountain trails and demonstrated on a quadrotor MAV [101,102]. Giusti et al. [101] casted the trail pattern recognition

problem as an image classification task: they developed trail perception technique based on Deep Neural Networks (DNNs), and trained and tested DNN using a large dataset acquired from a real-world human hiker [101]. The test results showed 95% of accuracy, which was better than the results classified by human (88–91%) and other algorithms (58–74%). However, the quadrotor often failed to perceive trails when there is not enough free space besides the trail centerline. Smolyanskiy et al. [102] introduced a DNN called TrailNet to follow a trail, which is inspired by Giusti et al. [101]. Training data were also collected from a human hiker, and the trained DNN was used to determine steering command. TrailNet DNN outperformed previous algorithms, showing smoother and more robust control [101,103].

Bartak et al. [104] proposed the machine learning algorithm to identify activities of flying UAVs from sensor reading [104]. They tried to enable UAVs flight autonomously without a controller from sensor reading based on machine learning. They designed four different approaches including hierarchical clustering (HIER), K-means clustering, Gaussian mixture model (GMM), and Hidden Markov Models (HMM). They estimated their applied machine learning algorithm in terms of accuracy and normalized mutual information using a flying UAV. However, this approach cannot correspond to multi-commands, and has low accuracy in different environment.

4. Discussion

Table 1 briefly summarizes the comparison of previous studies grouped by learning type (SL, UL, and RL), demonstration (simulation only, indoor test, and outdoor test), and application fields. In this survey, based on the studies on machine learning algorithm applied to UAVs for autonomous flight, the following issues are addressed;

- Machine learning is utilized for the control algorithm when it is required to find the optimal control gains of PID controller or weights of minimum cost function. Machine learning overcomes the nonlinearity and uncertainty of the controller by applying NNs or reinforcement learning methods to controller without additional computation in a random environment. The effectiveness of the parameter tuning was proved through simulations, whose results showed that Machine learning reduced computation, overshoot, and tracking error. However, most studies were limited to the virtual world in simulation or indoor environments, which means the controllers need to be tested experimentally with external disturbances such as wind and ground effect.
- Adaptive control methods make use of NN to process high-dimensional data. NN has shown great potential in complex maneuvers such as steep turn or aerobatic maneuver. Moreover, NN can reduce the training procedure compared to RL. Concurrent-learning adapts how to use current as well as stored data without affecting the responsiveness of the adaptive law to current data, and allows the use of direct training data, which improves the performance of the adaptive controller. RWNN was developed to combine the advantages of RNN and WNN, and it showed improved tracking performance. NN can be applied to dynamic model rather than just to the controller. In this case, the NN model is capable of learning the dynamics to a good accuracy in a complex system, especially in which it is hard to derive a physics-based model. However, the real world is full of uncertainty with massive data. Most studies are based on supervised learning, which trains existing control actions. For the complex control problems, it is required to aim at unsupervised learning methods. Furthermore, the test results focused on tracking performance, which were not enough to show autonomous control. Therefore, more investigation with other dynamic systems is required.
- Real-time path planning and navigation strategies have been studied for both single and multiple UAVs. For multi-agent path planning, reinforcement learning enables the cooperative plan to reduce unnecessary risks of task assignment and planning (iCCA), and parameter training helps balance between collision avoidance and the optimal path (CGLA). For a single UAV, RL, and ELM algorithm are helpful in finding an optimal path. The disaster sites are considered as reward states in RL, which the UAV is required to visit. The UAV can find a path to the target using only visual input or ascertain its positional information by capturing terrain data from the on-board sensor by using learning algorithms. Machine learning based on real-time path planning is a growing field in research of UAVs. For future work, the path planning and navigation strategies need to be tested on the finer and more complex grid, and eventually in the real world. Furthermore, the algorithms will be developed to find more efficient and fast paths and applied to the group of huge amount of UAVs.
- Object recognition based on machine learning has been utilized in UAVs for autonomous flight. UAVs based on machine learning are able to do designated tasks with collision avoidance, checking altitude and landing site, classification of crops, and so on. Q-value

Table 1. Comparison of learning algorithms grouped by learning type, demonstration, and application field.

	Algorithm	Learning type	Application field	Demonstration	Refs
Parameter tuning	FNN	SL	Attitude control	Simulation only	[50]
	RBF NN	SL	Attitude control	Simulation only	[51],[52]
	LA	RL	Trajectory tracking	Sim & Exp	[53],[54]
	LA	RL	Trajectory tracking	Simulation only	[55]
Adaptive control	FNN	SL	Aerobatic maneuver	Simulation only	[60]
	CL	SL	Trajectory tracking	Outdoor test	[61]
	CL,BKR-CL	SL	Transfer of controller	Indoor test	[62]
	RWNN	SL	Trajectory tracking	Simulation only	[63],[65]
	ReLU Network	SL	Aerobatic maneuver	Outdoor test	[66]
	NN	SL	Generalization of dynamics	Indoor test	[68]
	SVR	SL	Attitude control	Simulation only	[69]
	MDP	RL	Attitude control	Sim & Exp	[71]
	Q-learning	RL	Helicopter control	Simulation only	[72]
Path planning & navigation	CNAC,CSarsa	RL	Multi-UAV path	Simulation only	[77],[78]
	CGLA	RL	Single & multi-UAV path	Simulation only	[79]
	ACI	RL	Flocking control	Simulation only	[80]
	RL	RL	Single UAV path	Sim & Outdoor test	[81]
	ELM	SL	Single UAV navigation	Simulation only	[82]
	CNN	SL	Navigation in a city	Outdoor test	[83]
Collision avoidance	Q-learning	RL	Analysis of collision threat	Simulation only	[90]
	Imitation learning	SL	MAV flight through forest	Indoor & Outdoor test	[91]
	Q-learning	RL	Photo shoot of disaster site	Sim & Outdoor test	[81]
	CNN	SL	Trajectory prediction	Sim & Outdoor test	[92]
	NN,RL	SL,RL	Collision estimation	Sim & Outdoor test	[93]
	DNN,RL	SL,RL	Estimation of states	Indoor test	[94]
Object recognition on task	Regression	SSL	Estimation of altitude	Indoor test	[95]
	SVM	SL	Mapping a river	Outdoor test	[96]
	GMM,SVM	SL	landing site detection	Simulation only	[97]
	OBIA	SL	Precision agriculture	Sim & Outdoor test	[99]
	Hybrid deep NN	SL	Crop classification	Outdoor test	[100]
	DNN	SL	Mountain trail perception	Outdoor test	[101,102]
	K-means clustering, HIER,GMM,HMM	UL	Identification of UAV activities	Simulation only	[104]

algorithm was applied to analyze the threat of collision, and RL were applied to flight autonomously without collision. In addition, DNN, Gaussian Mixture Model, and Support Vector Machine were applied to predict the threat, estimate current status, and analyze surrounding environments from images. The UAVs are able to recognize object more efficiently with improvement. These approaches have proven to be very useful for object recognition. However, the demonstration of these approaches also depended on designated single task under limited environment like indoor navigation. For future work, studies are required to apply machine learning techniques stated above to UAVs under various outdoor environment for robustness and reliability against uncertainty and disturbance.

- Generally, deep networks and reinforcement learning require a large amount of data to train and teach algorithms. Like human brain, neural networks are required to experience a lot of data to ensure the algorithm to perform appropriately. RL also tends to need abundant data, and it is based on trial-and-errors, which means the algorithm needs to experience failures during training. These data-intensive algorithms make it difficult to use UAVs, which have battery problems and limited flight time. Furthermore, UAVs are often utilized to perform dangerous tasks on behalf of human. As long as machine learning algorithms rely on dataset collected in the past, they are exposed to dangerous situation for learning.
- In the past, researchers focused on enhancing control strategies by tuning parameters and reducing errors. Recently, as deep learning and reinforcement learning methods have been improved, and training data have been acquired from UAV flights, data-driven learning algorithms are spotlighted.

5. Conclusion

This paper described the state-of-the-art machine learning applied to UAVs for autonomous flight. All researchers in this research area have faced the issues regarding uncertain environment for autonomous flight. UAVs have challenges with highly nonlinear dynamics, obstacles, various patterns, and uncertainty in real world. As a result, machine learning has been studied for autonomous flight of UAVs including control strategy and object recognition. While these challenges continue to be addressed, a number of great advances have been made in the areas to UAVs with machine learning algorithms. UAVs are able to complete its task with stable

flight when the learning process is converged to satisfactory solution under uncertain environment. However, there are future directions in work related to UAVs. First, it is required to apply machine learning in all issues together for autonomous flight, not in specified issue. Furthermore, the algorithms should learn policies using small number of dataset, simulated data, or real-time data, rather than a large amount of data collected from real vehicles. Lastly, it is required to show reliability under more complicated situations in real world. It is required to demonstrate machine learning applied UAVs in real world to test its autonomous flight under uncertainty and disturbance.

Disclosure statement

No potential conflict of interest was reported by the authors.

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References

[1] Philip B. Domesticating the drone: the demilitarisation of unmanned aircraft for civil markets. Sci Eng Ethics. 2015;21(6): 1393–1412.
[2] Clarke R. Understanding the drone epidemic. Comput Law Secur Rev. 2014;30(3):230–246.
[3] Asaro PM. The labor of surveillance and bureaucratized killing: new subjectivities of military drone operators. Soc Semiotics. 2013;23(2):196–224.
[4] Kaminski ME. Drone federalism: civilian drones and the things they carry; 2013.
[5] Kristin BS, Kjersti L. The rise of the humanitarian drone: giving content to an emerging concept. Millennium. 2014;43(1):145–164.
[6] Chong J, Sweeney N. Civilian drone use in Canada; 2017.
[7] Daponte P, De Vito L, Lamonaca F, et al. Measurement science and education in the drone times. In 2017 IEEE International Instrumentation and Measurement Technology Conference (I2MTC); IEEE; 2017. p. 1–6.
[8] Kim P, Chen J, Kim J, et al. Slam-driven intelligent autonomous mobile robot navigation for construction applications. In Workshop of the European Group for Intelligent Computing in Engineering; Springer; 2018. p. 254–269.
[9] Zhao L, Chen J, Zhang Y, et al. Research on method of vision navigation for mobile robot in unstructured environment. In Optics, Photonics, and Digital Technologies for Imaging Applications V; International Society for Optics and Photonics; 2018. vol. 10679, p. 106791A.
[10] Kahn G, Villaflor A, Ding B, et al. Self-supervised deep reinforcement learning with generalized computation graphs for robot navigation. In 2018 IEEE International Conference on Robotics and Automation (ICRA); IEEE; 2018. p. 1–8.
[11] Luo Q, Duan H. Distributed uav flocking control based on homing pigeon hierarchical strategies. Aerosp Sci Technol. 2017;70:257–264.
[12] Liu C, Zhao X, Du Y, et al. Research on static path planning method of small obstacles for automatic navigation of agricultural machinery. IFAC-PapersOnLine. 2018;51(17):673–677.
[13] Marin-Plaza P, Hussein A, Martin D, et al. Global and local path planning study in a ros-based research platform for autonomous vehicles. J Adv Transp. 2018;2018.
[14] Papadakis P, Filliat D. Generic object discrimination for mobile assistive robots using projective light diffusion. In 2018 IEEE Winter Applications of Computer Vision Workshops (WACVW); IEEE; 2018. p. 60–68.
[15] Ye C, Qian X. 3-d object recognition of a robotic navigation aid for the visually impaired. IEEE Trans Neural Syst Rehabil Eng. 2018;26(2):441–450.
[16] Ruiz-Sarmiento J-R, Galindo C, Gonzalez-Jimenez J. Scene object recognition for mobile robots through semantic knowledge and probabilistic graphical models. Expert Syst Appl. 2015;42(22):8805–8816.
[17] High quality 3D laser ranging under general vehicle motion, Harrison, Alastair and Newman, Paul, IEEE International Conference on Robotics and Automation, 2008. ICRA 2008. IEEE; 2008. p. 7–12.
[18] Holz D, Droeschel D, Behnke S, et al. Fast 3d perception for collision avoidance and slam in domestic environments. In Mobile robots navigation. InTech; 2010.
[19] Shabbir J, Anwer T. A survey of deep learning techniques for mobile robot applications. arXiv preprint arXiv:1803.07608; 2018.
[20] Kim J, Canny J. Interpretable learning for self-driving cars by visualizing causal attention. In IEEE international conference on computer vision (ICCV); 2017.

[21] Xu H, Gao Y, Yu F, et al. End-to-end learning of driving models from large-scale video datasets. In (Proceedings of) IEEE international conference on computer vision and pattern recognition; 2017.
[22] Chen C, Seff A, Kornhauser A, et al. Deepdriving: learning affordance for direct perception in autonomous driving. In Proceedings of the IEEE International Conference on Computer Vision, 2015. p. 2722–2730.
[23] Zhang J, Springenberg JT, Boedecker J, et al. Deep reinforcement learning with successor features for navigation across similar environments. In 2017 IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS); IEEE; 2017. p. 2371–2378.
[24] Gan SK, Sukkarieh S. Multi-UAV target search using explicit decentralized gradient-based negotiation. In 2011 IEEE International Conference on Robotics and Automation (ICRA); IEEE; 2011. p. 751–756.
[25] Chelsea S, Kingston D, Cohen K. A formulation and heuristic approach to task allocation and routing of uavs under limited communication. Unmanned Syst. 2014;2(01):1–17.
[26] Bortoff SA. Path planning for UAVs. In (Proceedings of) the 2000 American Control Conference on; IEEE, 2000. vol. 1, p. 364–368.
[27] Meier L, Tanskanen P, Heng L, et al. Pixhawk: A micro aerial vehicle design for autonomous flight using onboard computer vision. Auton Robots. 2012;33(1–2): 21–39.
[28] Fraundorfer F, Heng L, Honegger D, et al. Vision-based autonomous mapping and exploration using a quadrotor mav. In 2012 IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS); IEEE, 2012. p. 4557–4564.
[29] Mancini M, Costante G, Valigi P, et al. Toward domain independence for learning-based monocular depth estimation. IEEE Robot Autom Lett. 2017;2(3):1778–1785.
[30] Oleynikova H, Honegger D, Pollefeys M. Reactive avoidance using embedded stereo vision for mav flight. In 2015 IEEE International Conference on Robotics and Automation (ICRA); IEEE; 2015. p. 50–56.
[31] Zhu Y, Mottaghi R, Kolve E, et al. Target-driven visual navigation in indoor scenes using deep reinforcement learning. In 2017 IEEE International Conference on Robotics and Automation (ICRA); IEEE, 2017. p. 3357–3364.
[32] Gandhi D, Pinto L, Gupta A. Learning to fly by crashing. In 2017 IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS); IEEE; 2017. p. 3948–3955.
[33] Sadeghi F, Levine S. Cad2rl: Real single-image flight without a single real image. Robot Sci Syst 13. 2017.
[34] Çelik K, Somani AK. Monocular vision slam for indoor aerial vehicles. J Electr Comput Eng. 2013;2013:4: 1573.
[35] Bills C, Chen J, Saxena A. Autonomous mav flight in indoor environments using single image perspective cues. In 2011 IEEE international conference on Robotics and automation (ICRA); IEEE, 2011. p. 5776–5783.
[36] KiKim D, Chen T. Deep neural network for real-time autonomous indoor navigation. arXiv preprint arXiv:1511.04668; 2015.
[37] AM Turing. Computing machinery and intelligence. Mind. 1950;59:433–460.l
[38] Moor J. The turing test: the elusive standard of artificial intelligence, vol. 30. New York: Springer Science & Business Media; 2003.
[39] Crevier D. AI: the tumultuous history of the search for artificial intelligence. New York: Basic Books; 1993.
[40] Pan Y. Heading toward artificial intelligence 2.0. Engineering. 2016;2(4):409–413.
[41] Russell SJ, Malik JM, Edwards DD. Artificial intelligence: a modern approach, vol. 2. Upper Saddle River, NJ: Prentice Hall; 2003.
[42] Lake BM, Ullman TD, Tenenbaum JB, et al. Building machines that learn and think like people. Behav Brain Sci. 2017;40.
[43] Tai L, Liu M. Deep-learning in mobile robotics-from perception to control systems: a survey on why and why not. arXiv preprint arXiv:1612.07139; 2016.
[44] Barber D. Bayesian reasoning and machine learning: Cambridge. Cambridge University Press; 2012.
[45] Kober J, Bagnell JA, Peters J. Reinforcement learning in robotics: A survey. Int J Rob Res. 2013;32(11):1238–1274.
[46] Sutton RS, Barto AG. Reinforcement learning: an introduction, vol. 1. Cambridge: MIT Press; 1998.
[47] Li Y. Deep reinforcement learning: an overview. arXiv preprint arXiv:1701.07274; 2017.
[48] Carrio A, Sampedro C, Rodriguez-Ramos A, et al. A review of deep learning methods and applications for unmanned aerial vehicles. J Sens. 2017;2017.
[49] LeCun Y, Bengio Y, Hinton G. Deep learning. Nature. 2015;521(7553):436.
[50] Efe MÖ. Neural network assisted computationally simple $PI^{\lambda}D^{\mu}$ control of a quadrotor UAV. IEEE Trans Ind Informat. 2011;7(2):354–361.
[51] Wang S, Li B, Geng Q. Research of rbf neural network pid control algorithm for longitudinal channel control of small UAV. In 2013 10th IEEE International Conference on Control and Automation (ICCA); IEEE; 2013. p. 1824–1827.
[52] Gao WN, Fan JL, Li YN. Research on neural network PID control algorithm for a quadrotor. Appl Mech Mater. 2015;719(0):346–351. Trans Tech Publ.
[53] Santos SRB, Givigi SN, Nascimento CL. Nonlinear tracking and aggressive maneuver controllers for quad-rotor robots using learning automata. In 2012 IEEE International Systems Conference (SysCon); IEEE, 2012. p. 1–8.
[54] Santos SRB, Givigi SN, Júnior CLN. An experimental validation of reinforcement learning applied to the position control of UAVs. In 2012 IEEE International Conference on Systems, Man, and Cybernetics (SMC); IEEE, 2012. p. 2796–2802.
[55] Jardine PT, Givigi S, Yousefi S. Parameter tuning for prediction-based quadcopter trajectory planning using learning automata. IFAC-PapersOnLine. 2017;50(1): 2341–2346.
[56] Schmidhuber. J. Deep learning in neural networks: an overview. Neural Netw. 2015;61:85–117.
[57] Goodfellow I, Bengio Y, Courville A, et al. Deep learning, vol. 1. Cambridge: MIT Press; 2016.

[58] Hinton GE, Osindero S, Teh Y-W. A fast learning algorithm for deep belief nets. Neural Comput. 2006;18(7):1527–1554.
[59] Wang H, Raj B, Xing EP. On the origin of deep learning. arXiv preprint arXiv:1702.07800; 2017.
[60] Kurnaz S, Cetin O, Kaynak O. Adaptive neuro-fuzzy inference system based autonomous flight control of unmanned air vehicles. Expert Syst Appl. 2010;37(2): 1229–1234.
[61] Chowdhary GV, Johnson EN. Theory and flight-test validation of a concurrent-learning adaptive controller. J Guid Control Dyn. 2011;34(2):592–607.
[62] Chowdhary G, Wu T, Cutler M, et al. Rapid transfer of controllers between UAVs using learning-based adaptive control. In 2013 IEEE International Conference on Robotics and Automation (ICRA); IEEE; 2013. p. 5409–5416.
[63] Lin C-J, Chin C-C. Prediction and identification using wavelet-based recurrent fuzzy neural networks. IEEE Trans Syst Man Cybern Part B (Cybern). 2004;34(5): 2144–2154.
[64] Yoo SJ, Park JB, Choi YH. Adaptive dynamic surface control of flexible-joint robots using self-recurrent wavelet neural networks. IEEE Trans Syst Man Cybern Part B (Cybern). 2006;36(6):1342–1355.
[65] Lin C-M, Tai C-F, Chung C-C. Intelligent control system design for UAV using a recurrent wavelet neural network. Neural Comput Appl. 2014;24(2): 487–496.
[66] Punjani A, Abbeel P. Deep learning helicopter dynamics models. In 2015 IEEE International Conference on Robotics and Automation (ICRA); IEEE, 2015. p. 3223–3230.
[67] Abbeel P, Coates A, Ng AY. Autonomous helicopter aerobatics through apprenticeship learning. Int J Rob Res. 2010;29(13):1608–1639.
[68] Bansal S, Akametalu AK, Jiang FJ, et al. Learning quadrotor dynamics using neural network for flight control. In 2016 IEEE 55th Conference on Decision and Control (CDC); IEEE; 2016. p. 4653–4660.
[69] Shin J, Kim HJ, Kim Y. Adaptive support vector regression for UAV flight control. Neural Netw. 2011;24(1): 109–120.
[70] Powell WB. AI, OR and control theory: A rosetta stone for stochastic optimization. Princeton (NJ): Princeton University; 2012.
[71] Bou-Ammar H, Voos H, Ertel W. Controller design for quadrotor uavs using reinforcement learning. In 2010 IEEE International Conference on Control Applications (CCA); IEEE; 2010. p. 2130–2135.
[72] Lee D, Choi M, Bang H. Model-free linear quadratic tracking control for unmanned helicopters using reinforcement learning. In 2011 5th International Conference on Automation, Robotics and Applications (ICARA); IEEE; 2011. p. 19–22.
[73] Lee D, Kim S, Suk J. Formation flight of unmanned aerial vehicles using track guidance. Aerosp Sci Technol. 2018;76:412–420.
[74] Ambroziak L, Gosiewski Z. Two stage switching control for autonomous formation flight of unmanned aerial vehicles. Aerosp Sci Technol. 2015;46:221–226.
[75] Redding J, Geramifard A, Undurti A, et al. An intelligent cooperative control architecture. In American Control Conference (ACC), 2010; IEEE; 2010. p. 57–62.
[76] Redding J, Geramifard A, How JP. Actor-critic policy learning in cooperative planning. In AAAI Spring Symposium: Embedded Reasoning, 2010.
[77] Geramifard A, Redding J, Roy N, et al. UAV cooperative control with stochastic risk models. In American Control Conference (ACC), 2011; IEEE; 2011. p. 3393–3398.
[78] Geramifard A, Redding J, How JP. Intelligent cooperative control architecture: a framework for performance improvement using safe learning. J Intell Robot Syst. 2013;72(1):83–103.
[79] Zhang B, Liu W, Mao Z, et al. Cooperative and geometric learning algorithm (CGLA) for path planning of UAVs with limited information. Automatica. 2014;50(3):809–820.
[80] Xu H, Carrillo LRG. Fast reinforcement learning based distributed optimal flocking control and network co-design for uncertain networked multi-UAV system. In Unmanned Systems Technology XIX; International Society for Optics and Photonics; 2017. vol. 10195, p. 1019511.
[81] Junell JL, Van Kampen E-J, de Visser CC, et al. Reinforcement learning applied to a quadrotor guidance law in autonomous flight. In AIAA Guidance, Navigation, and Control Conference; 2015. p. 1990.
[82] Kan EM, Lim MH, Ong YS, et al. Extreme learning machine terrain-based navigation for unmanned aerial vehicles. Neural Comput Appl. 2013;22(3–4):469–477.
[83] Loquercio A, Maqueda AI, delBlanco CR, et al. Dronet: Learning to fly by driving. IEEE Robot Automat Lett. 2018;3(2):1088–1095.
[84] Vandapel N, Kuffner J, Amidi O. Planning 3-d path networks in unstructured environments. In (Proceedings of) the 2005 IEEE International Conference on Robotics and Automation (ICRA); IEEE; 2005. p. 4624–4629.
[85] Bachrach A, He R, Roy N. Autonomous flight in unknown indoor environments. Int J Micro Air Vehicle. 2009;1(4):217–228.
[86] Bry A, Bachrach A, Roy N. State estimation for aggressive flight in GPS-denied environments using onboard sensing. In 2012 IEEE International Conference on Robotics and Automation (ICRA); IEEE; 2012. p. 1–8.
[87] Bachrach A, Prentice S, He R, et al. Estimation, planning, and mapping for autonomous flight using an RGB-D camera in GPS-denied environments. Int J Rob Res. 2012;31(11):1320–1343.
[88] Achtelik M, Achtelik M, Weiss S, et al. Onboard IMU and monocular vision based control for mavs in unknown in-and outdoor environments. In 2011 IEEE international conference on Robotics and automation (ICRA); IEEE; 2011. p. 3056–3063.
[89] Wendel A, Maurer M, Graber G, et al. Dense reconstruction on-the-fly. In 2012 IEEE Conference on Computer Vision and Pattern Recognition (CVPR); IEEE; 2012. p. 1450–1457.
[90] Lei G, Dong M-z, Xu T, et al. Multi-agent path planning for unmanned aerial vehicle based on threats analysis. In 2011 3rd International Workshop on Intelligent Systems and Applications (ISA); IEEE; 2011. p. 1–4.

[91] Ross S, Melik-Barkhudarov N, Shankar KS, et al. Learning monocular reactive uav control in cluttered natural environments. In 2013 IEEE international conference on robotics and automation; IEEE; 2013. p. 1765–1772.
[92] Yang S, Konam S, Ma C, et al. Obstacle avoidance through deep networks based intermediate perception. arXiv preprint arXiv:1704.08759; 2017.
[93] Kahn G, Villaflor A, Pong V, et al. Uncertainty-aware reinforcement learning for collision avoidance. arXiv preprint arXiv:1702.01182; 2017.
[94] Zhang T, Kahn G, Levine S, et al. Learning deep control policies for autonomous aerial vehicles with mpc-guided policy search. In 2016 IEEE international conference on robotics and automation (ICRA); IEEE; 2016. p. 528–535.
[95] Cherian A, Andersh J, Morellas V, et al. Autonomous altitude estimation of a UAV using a single onboard camera. In IEEE/RSJ International Conference on Intelligent Robots and Systems, 2009. IROS 2009; IEEE; 2009. p. 3900–3905.
[96] Scherer S, Rehder J, Achar S, et al. River mapping from a flying robot: state estimation, river detection, and obstacle mapping. Auton Robots. 2012;33(1–2):189–214.
[97] Guo X, Denman S, Fookes C, et al. Automatic UAV forced landing site detection using machine learning. In 2014 International Conference on Digital lmage Computing: Techniques and Applications (DlCTA); IEEE; 2014. p. 1–7.
[98] Mejias L, Fitzgerald D, Eng P, et al. Forced landing technologies for unmanned aerial vehicles: towards safer operations. In Aerial vehicles. InTech; 2009.
[99] Pérez-Ortiz M, Peña JM, Gutiérrez PA, et al. Selecting patterns and features for between-and within-crop-row weed mapping using UAV-imagery. Expert Syst Appl. 2016;47:85–94.
[100] Rebetez J, Satizábal HF, Mota M, et al. Augmenting a convolutional neural network with local histograms: a case study in crop classification from high-resolution UAV imagery. In European Symp. on Artificial Neural Networks, Computational Intelligence and Machine Learning; 2016. p. 515–520.
[101] Giusti A, Guzzi J, Ciresan DC, et al. A machine learning approach to visual perception of forest trails for mobile robots. IEEE Rob Autom Lett. 2016;1(2):661–667.
[102] Smolyanskiy N, Kamenev A, Smith J, et al. Toward low-flying autonomous mav trail navigation using deep neural networks for environmental awareness. In IEEE/RSJ International conference on intelligent robots (IROS); 2017.
[103] He K, Zhang X, Ren S, et al. Deep residual learning for image recognition. In Proceedings of the IEEE conference on computer vision and pattern recognition; 2016. p. 770–778.
[104] Barták R, Vomlelová M. Using machine learning to identify activities of a flying drone from sensor readings. In (Proceedings of) the Thirtieth International Florida Artificial Intelligence Research Society Conference, FLAIRS; 2017. p. 436–441.